Lean proof simplification after the actual Wang row commits

A third review assuming all earlier proposals are available

Proof-source commit `22e020937caa81d0cb72a4f63071d6c0bcc1eba9`

1 Keep the small deletion and price stronger interfaces honestly

After the final novelty check, the immediate source compression is five physical lines in one module. Two new wall-bound proofs can use their stored finite bounds and monotonicity rather than arithmetic enumeration. No helper or import is needed. Two stronger row interfaces also survive the comparison, but neither currently shortens the complete affected proof source.

Candidate	Additional obligation or deletion	Recommendation
Direct wall bounds	Use `Fin 3` bounds and natural-power monotonicity	First pilot; net −5 physical lines.
Receiver causality	Every actual receive tile lies at the actual moved head, without temporal matching	Optional API; net +17 proof lines.
Complete tile coherence	An extracted interior tile equals the independently constructed `stepTile`	Optional API; net +3 proof lines.

The late comparison materially reduced the recommendation set. Incoming material already proposes the exact absolute-value trace cleanup and bounded first-failure abstraction; both are withdrawn. The latest maintained source already proves marked-word halting through the periodic-origin color contradiction; that endpoint is not a new suggestion either. Their patches and savings are not part of this report's recommendations.

All previous ideas are assumed available. The original ledger contains 60 initial proposal families, 28 later families and five previous-round deltas: 93 assumed contracts. The final comparison additionally covers the later material at the comparison pin below. Generic constructor views, direction descriptors, coherent facts, shared-edge row construction, carry normalization, scanner iteration, DAG invariants and canonicality strategies receive no new credit. An assumed contract is not a claim that every alternative design coexists in one compiled tree.

Two snapshots serve different purposes. Exact code examples and metrics use `22e020937caa81d0cb72a4f63071d6c0bcc1eba9`. The bounded final source/corpus comparison uses `1e2a853935a6155cbcf2aa199aaca51946c4fff9`. The retained target files are byte-identical at the two pins. The final corpus has 21 manuscripts: 18 placed reports and three incoming packets. The initial 383-section ledger is not presented as a full late-corpus census.

Validation boundary. The archive contains replacement candidates, patches, source and finite-model audits, and source-pinned evidence. **Every new Lean candidate remains uncompiled and not kernel-checked.** Patch checks and finite tests do not establish elaboration, imported theorem resolution, axiom footprints or universal proofs. Compile with the pinned project before adoption.

The useful result is narrower than a large compression campaign: one low-cost deletion and two exact extraction-side contracts whose benefit is reuse. No compiler speedup, consistency assertion, or full polynomial/internal-ZFC equivalence is claimed.

2 Recent commits and the newer maintained endpoint

From the preceding review pin `b310142330df89e48413c8fce62e7de223e1c7d5` to the proof-source pin, these eight mathematical commits add 32 main-directory Lean files: 16 theorem modules with 2,099 physical lines and 16 check modules with 1,224, totaling 3,323. These are counts of the selected sequence, not every repository change. Exact paths and hashes are in `catalog/new_machine_commits.json`.

Commit	New interface relevant to the retained proposals
`72d3198bb`	Finite initialization walls and exact horizontal reset alternatives.
`076b953e7`	Persistent walls and a seam-free open strip at every integer time.
`90a2ded65`	Actual first step and exact configuration-color semantics.
`edf3a19d0`	Exact tile-input grammar; one finish event propagates to a finish/reset row.
`218631436`	Actual clock-one row; no premature finish in a live encoded row.
`993d6730a`	Actual signals, exact receiver targets and operational color updates.
`04267333b`	Strict strip head containment and sharp blank-run trace bounds.
`a421ec86e`	Actual row evolution and an independently constructed matching step row.

The last commit provides two distinct sides: `row_step` extracts the actual step of an encoded configuration, while `stepTile` constructs a row from it. Their overlap supports the narrow complete-tile proposal; it does not make a generic simulation framework new. See `WangTiles942RowStep.lean:164–203` and `WangTiles942StepConstruction.lean:22–44`.

The later maintained source goes further. Commit `512e0f6a3` adds the actual marked-word halting theorem, using existing row evolution and the origin-color contradiction, and `CurrentSoundness2011` proves positive polynomial zero ⇒ actual blank halt ⇒ syntactic ZFC refutation. Hence ConZFC excludes positive zeros of the maintained polynomial and three thirteen-variable forms. See `WangTiles942Soundness.lean:24–68 (late)` and `CurrentSoundness2011.lean:24–70 (late)`.

Commit `be45083ae` also closes the marked-language equivalence: existence of an actual marked word is equivalent to actual blank halting; absence of marked words is equivalent to ConZFC. The cofinal construction admits independently large width and height. These are maintained external Lean statements, not new proposals here; see `WangTiles942Completeness.lean:28–79 (late)`. The actual polynomial converse and internal-ZFC formalization are still separate obligations.

Earlier adoption is not new credit. Nine groups already remove 264 maintained lines, and the total-degree tactic is maintained. At the proof-source pin, some queued scanner, model-axiom and arithmetic edits were still absent despite adoption prose. All are nevertheless treated as available for novelty. The report does not reuse their savings, mistake queue status for integration, or describe the now-maintained halting endpoint as future work.

3 The final exclusion screen and the surviving distinction

The baseline is a comparison of contracts, not an invented merged implementation. Earlier proposals include alternatives, sketches and dependencies rather than one compatible patch set. Assuming them available means each retained suggestion must add a specific conclusion or simplify a newly introduced client beyond those ideas. It does not permit a fictitious post-adoption line count.

The initial corpus audit covers 11 actual manuscripts and 383 section records, with 93 assumed contracts. A bounded final pass at `1e2a853935a6155cbcf2aa199aaca51946c4fff9` broadens the comparison to **21 manuscripts**, comprising 18 placed reports and three incoming packets. The late addendum is separate from the original section ledger. Later placement does not make an old manuscript new, and incoming suggestions count whether implemented, deferred or merely sketched.

Retained candidate	Assumed earlier work	Exact additional conclusion
Receiver causality	LE10 constructor views, E19 directions, R2N4 fibers; later shared-edge construction	A receive tuple in an *actual* encoded row must occupy the actual moved-head coordinate, without temporal matching.
Complete tile coherence	LE08 canonicality, LE10 views, R2N4/R2N5 fibers/coherence; later shared-edge construction	An arbitrary actual allowed/matching encoded row equals the independent constructor inside the strip.
Direct wall bounds	Existing library facts and earlier local-cleanup proposals	Delete the exact new finite-bound/enumeration tails, without claiming a new proof pattern.

The shared-edge proposal is already assumed applied. The later structural-refinements packet proposes equating the *constructed* `stepTile` with an edge-built row, making horizontal matching structural. This report does not propose that representation or its one-bridge geometry again. The retained contracts instead concern an arbitrary actual row T satisfying tile, matching and encoding constraints. A convenient representation of the constructor alone does not rule out an unwanted receiver in T or identify all its interior tiles.

Withdrawn rather than renamed. T1's first-exit section and T3's first-failure section cover the same initial-wall search clients; their generalization is excluded. T3 also gives the exact absolute-value corollary replacement, so its five-line patch is excluded. The periodic-origin endpoint is already in maintained source. The final delivery claims no novelty or saving for these candidates. The late ledger records packet identities and precise source ranges, including T1's `report/sections/05_first_exit.tex`:1–89 and T3's `report.tex`:320–352 (first failure), 353–439 (shared edges), and 440–449 (trace).

The 93-contract source ledger still supplies the earlier dependencies, including generic iteration, coherent facts, finite views, directions and canonicality. Since their hypothetical implementations may already shorten parts of the current clients, the figures below are *pinned-source measurements only.* The defensible new contribution is the causal or extraction/construction conclusion, not an unmeasured post-baseline compression claim.

4 Direct wall bounds and complete cost accounting

The remaining immediate deletion needs no helper. In `WangTiles942HeadBounds.lean:59–74`, the seam parameters already lie in `Fin 3`. After the existing tuple extraction and rewrite, the left wall needs only $b \le 2$ and the right wall needs $a^2 \le 4$. Replace their remaining tails with:

```
-- left_wall_east_le_two, after obtain and rw [h]
  exact Nat.le_of_lt_succ b.isLt

-- right_wall_west_le_four, after obtain and rw [h]
  exact Nat.pow_le_pow_left (Nat.le_of_lt_succ a.isLt) 2
```

The second proof converts the stored $a < 3$ into $a \le 2$, then applies natural-power monotonicity; 2^2 reduces to 4. Existing uses in `SidonPositions.lean:33–35` and `NativePellPower.lean:70–72` confirm the argument order. The complete module changes from **135 to 130 physical lines**, removing 117 UTF-8 bytes. All nine public headers remain unchanged.

The wall asymmetry remains essential: the left-facing bound is 2, the right-facing bound is 4, and an active transition rail is at least 6. No symmetry of the full tile language is introduced and neither proof claims both transition edges are active. This is a small concrete cleanup, not architecture-level novelty.

Candidate and measurement	Scope	Before / after	Delta
Wall bounds	Whole file, physical	135 / 130	−5
Receiver causality	Affected proof bodies	47 / 64	+17
Receiver causality	Whole file, physical	205 / 252	+47
Complete tile coherence	Affected proof bodies	38 / 41	+3
Complete tile coherence	Whole file, physical	205 / 233	+28
Complete tile coherence	Whole file, nonblank	191 / 217	+26
Both row interfaces	Whole file, physical	205 / 280	+75

Physical lines include blanks and comments; nonblank lines retain comments but omit blanks; affected proof-body counts include all new helpers. The separately composed row candidate has +20 affected proof-body lines and +71 nonblank lines. Its unified patch must be used if both interfaces are wanted: the standalone patches have overlapping contexts and do not apply sequentially in either order.

These are measured changes, not a promise about an unknown tree implementing all earlier proposals. The complete-tile file also grows by 1,275 bytes; the combined row file grows by 3,199 bytes. No compiler timing, heartbeat, proof-term-size or compiled-size measurements were made. Shorter wrappers alone are not a source-size win when their moved proof and helper bodies are included.

Adopt the direct wall patch first after compilation. Treat either row interface as a deliberate semantic investment, justified by its exact downstream consumer rather than attractive client-only counts. The excluded trace and first-failure patches contribute zero to the final saving.

5 An optional causal statement for actual receive tiles

Move the spatial exclusion into one reusable theorem only if its conclusion has value to later clients. The current `far_west_zero` and `far_east_zero` proofs independently inspect the neighboring wall, plain tile or head transition. A stronger statement packages the reason behind both arguments: every receive tile inside a live encoded row is located exactly at the actual moved head.

Let $p = \text{cfg.head}$, let a be the actual transition at p, and let $n = a.\text{move.apply}(p)$. The proposed `receive_at_next_head` assumes allowed tiles, horizontal matching, `PersistentStrip`, the original `RowEncodes`, a live state, and an actual directional receive tuple at an interior position x. It concludes $x = n$. It has **no temporal-matching or no-finish hypothesis**. Its target assumption is membership in `targets d`, which retains target 215.

The argument is spatial.

1. A receive tile has a nonzero incoming rail, at least 6. Inspect its incoming-side neighbor, one integer cell away.
2. That neighbor is either interior or exactly the wall. A wall is impossible because its facing edge is at most 2 on the left or 4 on the right.
3. If it is an interior non-head cell, both north inputs are plain. The existing `plain_pair_edge_zero` forces the matching rail to zero, a contradiction.
4. The neighbor is therefore the actual head. Its forced transition has one inactive zero edge. Matching the positive rail identifies the move direction, and the adjacent-coordinate equation gives $x = n$.

This proof needs positivity and the inactive edge, not rail injectivity. Exact target identification remains owned by the existing directional receiver theorems; it is not claimed as an additional result of this helper.

The second helper, `far_pass_or_finish`, eliminates only the receive case of the exact plain-input grammar. Away from p and n, with b the original tape bit, it concludes

$$T(x,t) = \langle 0, b, 0, b\rangle \quad \vee \quad T(x,t) = \langle 0, b, 0, 438\rangle.$$

Both zero-edge clients become projections of this disjunction. Only `far_pass` invokes the independently derived `no_south438` result to remove the finish alternative. This keeps the temporal premise exactly where it belongs.

Cost and decision. The two edge proofs shrink from 22 and 21 proof lines to two each. But the new causal and disjunction proofs cost 49 and 7 lines. Including the unchanged four-line `far_pass`, the affected bodies grow from **47 to 64 lines**; the full file grows from **205 to 252 physical lines**. Thus this is an optional semantic API, not a current net-shortening recommendation. Prefer it only when future spatial-only consumers need receiver location, or when centralizing this invariant is worth the source cost.

The seven existing exported statements remain unchanged. The supplied candidate is concrete Lean source, not pseudocode that assumes a future direction API. Its sources are `WangTiles942RowStep.lean:35–122`, `WangTiles942Signals.lean:54–104`, and the full new declarations in `row/helper.lean`. The novelty is this exact row-level causal conclusion; constructor views and direction abstraction are already credited to the baseline.

6 An optional equality between extraction and construction

Preserve complete tuple equalities before projecting south colors. The current receiver proof obtains an exact receiving tile in each direction, then immediately discards three edges with `congrArg Tile.south`. Later, `south_step` repeats the operational old-head, moved-head and unchanged-cell partition. The reverse construction already owns that same partition in `stepTile`.

The proposed new conclusion is deliberately restricted to the open strip:

$$-L < x < R \quad \Longrightarrow \quad T(x,t) = \mathrm{stepTile}(\mathrm{cfg}, q, x).$$

Its hypotheses are the original `south_step` hypotheses: allowedness, horizontal and temporal matching, persistent walls, the same encoded configuration, and its live state. There is no replacement configuration and no added premise that the successor remains live.

Separate the two proof responsibilities. The actual-row argument owns identification of the full tile. The independent constructor owns the operational read/write/move interpretation of that tile. The equality proof has only three cases:

1. At the old head, `head_transition` identifies the literal cached transition tile, the first branch of `stepTile`.
2. At the moved head, a new `receiver_tile` retains the complete equality already obtained in the old `receiver_south` proof. Simplification matches the constructor's `receiveTile`.
3. Everywhere else in the interior, the existing `far_pass` identifies the exact pass tuple. The original temporal no-finish reasoning is still required there.

Then `south_step` becomes a projection followed by the independent constructor theorem:

```
exact (congrArg Tile.south
  (row_tile_eq_stepTile hA hH hTime hs hrow q hq x hl hr)).trans
  (stepTile_south cfg q hq x)
```

The supplied candidate adds the StepConstruction import; the inspected import graph has no path back to RowStep. `receiver_south` remains as a compatibility projection, and `row_step` retains its existing statement and proof.

Do not mistake moved proof text for deleted proof text. The old receiver and south proofs total 38 nonblank body lines; their new wrappers total seven. The two new proofs cost 34, making **41 lines overall, a net increase of three**. The complete file grows from 205 to 233 physical lines; nonblank lines rise from 191 to 217, and bytes from 9,481 to 10,756. The case for adoption is an exact semantic boundary, not a claimed 31-line saving.

The same equality does *not* identify the seam tiles, exterior plane, complete configurations or tile indices. The source observation interval is explicit in `WangTiles942RowEncoding.lean:21–26`. The receiver/source anchors are `WangTiles942RowStep.lean:124–186`, the constructor at `WangTiles942StepConstruction.lean:22–44`, and its independent south theorem at `WangTiles942StepConstruction.lean:98–114`.

This patch and receiver causality both edit RowStep. Use the separately supplied combined patch rather than applying the standalone patches sequentially: their unified-diff contexts overlap. The canonical candidate can consume the unchanged `far_pass`; it does not require the causal helper.

7 Preserve the semantic guards that make the proofs true

Entry into halt is still an active step. A live transition with next state `none` writes at the old head and moves once. At the destination it reads the *original destination bit*, because the write occurred at the distinct old head. Receive target 215 is valid and yields south color 432 or 433; it must not be rejected as an out-of-range operating state. The maintained statement is `WangMachineStep.lean:115–123`, and both proposed row interfaces retain these cases.

An unchanged plain cell can locally finish. The tuple $\langle 0, 0, 0, 438\rangle$ is a genuine finish alternative with plain north and zero horizontal edges. Those conditions alone therefore do not imply pass. Receiver causality deliberately yields pass *or* finish. The stronger pass conclusion uses `no_south438`, whose proof depends on temporal matching and the live head. A finite local row containing a head, its halt receiver and a separate finish illustrates the forbidden local strengthening; it is not claimed to satisfy every persistent-strip or temporal premise.

Do not strengthen observation to configuration equality. In the interval $(-2, 3)$, configurations with state `some 1`, head 1, and tapes `[0]` versus `[0,100]` have identical observed colors but different tapes. Even equality of all integer reads does not imply equality of the concrete tape lists: `[0]` and `[0,0]` are observationally equal and list-unequal. The complete-tile proposal proves tuple equality only at observed interior positions. It cannot recover exterior tape content, a canonical tape representation or tile-index injectivity.

The initializer is not a blank configuration. The actual clock-one configuration has tape `[0]`, head 1 and state `some 1`; its colors are 1 at zero, 4 at one, and 0 elsewhere. See `DecimalMachineInitialization.lean:20–52` and `WangMachineEncoding.lean:147–169`. The head initializer is the literal $\langle 3, 436, 4, 4\rangle$, not a similar tuple with south 188, as specified in `WangTiles942InitialStrip.lean:23–34`. No proposal replaces this with an arbitrary initial state or independently selected witness.

Keep spatial and temporal dependencies separate. Receiver location needs allowed tiles, horizontal matching, persistent walls, the original encoded row and its live state. It does not need temporal matching. Complete tile equality does need the temporal information that rules out an unchanged finish. Moving both results into one stronger context would hide this useful distinction and make the causal theorem unnecessarily difficult to reuse.

All positions remain signed integers. The actual destination is the one-cell move of the old head, not a natural-index subtraction. Wall-edge bounds retain their asymmetric values, and no statement extends the interior equality to seam walls or exterior cells. The original destination tape bit remains unconstrained; no blank-tape premise is added to a general row theorem.

Respect the newer endpoint without overstating it. The late maintained source already derives marked-language consistency equivalence and the forward consistency-to-polynomial-unsolvability implication. The retained local proposals do not reprove those results and are not prerequisites for them. They also do not provide the remaining reverse arithmetic/controller construction or an internal-ZFC formalization. Their scope is proof maintenance and stronger actual-row interfaces only.

8 What was checked and what still requires Lean

Static evidence is reproducible but deliberately weaker than a proof check. The retained wall patch preserves all nine public headers in HeadBounds. Each RowStep alternative and the combined candidate preserve its seven existing exported statements; unrelated proof bodies remain unchanged. The canonical audit checks that its added import creates no cycle back to RowStep and introduces no `sorry`, `admit`, `axiom` or `native_decide`. Source hashes bind each audit to the original pin, and the final comparison confirms retained target-source identity at the later pin.

Independent finite checks cover these concrete semantic surfaces:

- Causality: all 430 cached actions, 488 receive tuples, 242,064 plain/plain pairs, 4,392 receive/seam incoming interfaces, and 209,840 action/receiver interfaces. The 37,788 matching plain pairs have zero matching edge; 860 actual action/receiver matches remain, including both halt-target receiver cases.
- Canonical row: 82,560 local configuration/tape cases and 577,920 color/horizontal point checks, including 192 entering-halt cases; 6,880 five-cell rows under the finite test's explicit zero-boundary and no-finish restrictions. Those restrictions are test-model conditions, not new production premises.
- Composition: the standalone RowStep patches fail sequential application in both orders because their contexts overlap. The supplied combined patch applies to the original source and agrees exactly with the composed candidate, retaining all existing statements.

These checks do not validate the universal Lean proof, coercion inference, exact simplifier behavior or an axiom footprint. Original upstream check receipts apply to original code only. The portable audits use explicit failures rather than relying on removable Python `assert` statements; ordinary and optimized-mode results and rejection controls are recorded separately.

Replay the retained source checks. From the extracted archive root, with its supplied frozen sources, run:

```
python3 strip/verify_static.py
python3 row/audit_receiver_causality.py
python3 row/audit_combined_row.py
python3 machine/audit.py
```

These audits print JSON to stdout by default; output files require an explicit option. Row/machine audits use the supplied `baseline/`; the wall verifier uses its own source snapshot. The catalog verification has separate corpus-input requirements. No compiler is installed and no patch is applied to the original checkout by these commands.

The remaining adoption gate. Use a disposable checkout with the pinned project toolchain. Compile the wall patch and its existing HeadBounds checks first. Trial each RowStep alternative independently, or use the supplied composition. Compile RowStep, StepConstruction, existing check modules and supplied prototype checks; inspect `#print axioms` and rebuild reverse dependencies, including the newer maintained soundness consumers. Check the receiver tuple `simpa` and definitional `nextHead`/`sourceIndex` reductions early.

Exact blocker. Target: changed Lean modules and downstream checks in the pinned Lean 4.32.0 project. Action: elaboration, kernel checking and axiom inspection. Reason: no Lean executable is available in this review environment. Status: **not run**. No compiler success or failure is inferred; the new proof texts remain proposals.

9 Evidence map and adoption order

Start with the smallest bounded delta. Compile the wall-bound replacement first. Receiver causality and complete-tile coherence each deserve an independent decision: do their exact conclusions have an actual downstream consumer worth the current positive source cost? If both are wanted, use the supplied combined patch rather than applying the two standalone patches sequentially.

Archive area	What it establishes
`catalog/`	Initial 93-contract/383-section ledger; original manuscript identity/adoption evidence; supplemental late-corpus and withdrawal decisions.
`strip/`	Retained HeadBounds patch and candidate, preserved-header/source-diff checks and exact size measurements.
`row/`	Receiver-location helper, complete alternative RowStep candidate, measured composition, patches and semantic negative controls.
`machine/`	Complete-tile alternative, independent finite audit, complete source/body costs and interior-only scope checks.
`report/`	Editable English TeX, rendered PDF and a rebuild script using installed TeX files without downloads.

Original corpus provenance is preserved separately. At the proof-source pin, the six placed packets account for 1,866 original member records: 296 byte-identical files, 14 CSVs changed only from CRLF to LF, 1,513 omitted members verified against 818 distinct identical Git blobs, 41 ignored logs and two ignored checksum files. Five later ZIPs add 400 members and match the earlier late corpus exactly. These figures audit the original eleven manuscripts; they are not totals for the final 21-manuscript comparison.

The newest corpus pass covers subsequent placements and incoming manuscripts at the comparison pin, and supersedes earlier positive novelty judgments where necessary. The important practical result is the withdrawal of first-failure sharing, the exact absolute-value cleanup and the already-maintained periodic halting endpoint. Old hashes and decisions remain evidence of review history, not authority to restore a withdrawn recommendation.

The core provenance records are `catalog/corpus_validation.json`, `catalog/corpus_member_hashes.json`, `catalog/adoption_evidence.json`, `catalog/assumed_contracts.json` and `catalog/late1e2/late_exclusion_ledger.json`. They distinguish implementation facts from assumed contracts. Corpus inspection does not treat old supplied compiler logs as validation of new code or execute old packet scripts.

Pinned links, not moving branch locations. Ordinary source links in this report use `22e020937caa81d0cb72a4f63071d6c0bcc1eba9`; links marked “late” use `1e2a853935a6155cbcf2aa199aaca51946c4fff9`. Retained code targets are identical at these pins. Before adoption on another checkout, compare source hashes and rebuild the relevant checks rather than assuming that unchanged theorem names imply identical definitions or imports.

Final decision. After assuming the old work and honoring the late exclusions, this review does not support another large automatic compression pass. It supports five low-overhead lines to remove and two narrowly scoped extraction-side contracts whose value is reuse. The next concrete result to establish is compilation of the small wall patch; stronger row interfaces should follow only when their consumers justify them.